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Bias-Free Langua

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Introduction

This document describes how to upload files to Cisco Technical Assistance Center (TAC).

Overview

The TAC customer support engineers can assist you with resolving an issue in a timely manner when they have relevant files attached to the issue. You have several options to upload the files related to your issue. Some of these options are less secure, and can lead to certain inherent risks, and each option has limitations that you need to consider before deciding on an appropriate upload option. Table 1 summarizes the available upload options with details on file encryption capabilities, recommended files size limits, and other relevant information.

Table 1. Available Upload Options

Available Option (In Order of Preference)		Files are Encrypted in Transit	Files are Encrypted at Rest	Recommended File Size Limit
Support Case Manager (SCM)	How To	Yes	Yes	No limit
Customer eXperience Drive	How To	Yes*	Yes	No limit
Email to attach@cisco.com	How To	No**	Yes	20 MB or less based on customer mail server limits
*Applies to all protocols and FTP is no longer supported. **You must encrypt prior to transit. Secure transit is guaranteed only from the point where the email/attachment reaches the Cisco network, not on the customer network or email provider side.				

Support Case Manager File Upload

The Support Case Manager (SCM) file upload method is a secure option for uploading files to cases. The communication channel between your computing device and Cisco is encrypted. Files uploaded through SCM are immediately linked to the associated case and stored in an encrypted format.

Uploading a File to a Case

After you submit the case, you can upload files.

Step 1. Log in to SCM.

Step 2. In order to view and edit the case, click the case number or case title in the list. The Case Summary page opens.

Step 3. Click **Add Files** in order to choose a file and upload it as an attachment to the case. The system displays the SCM File Uploader tool.

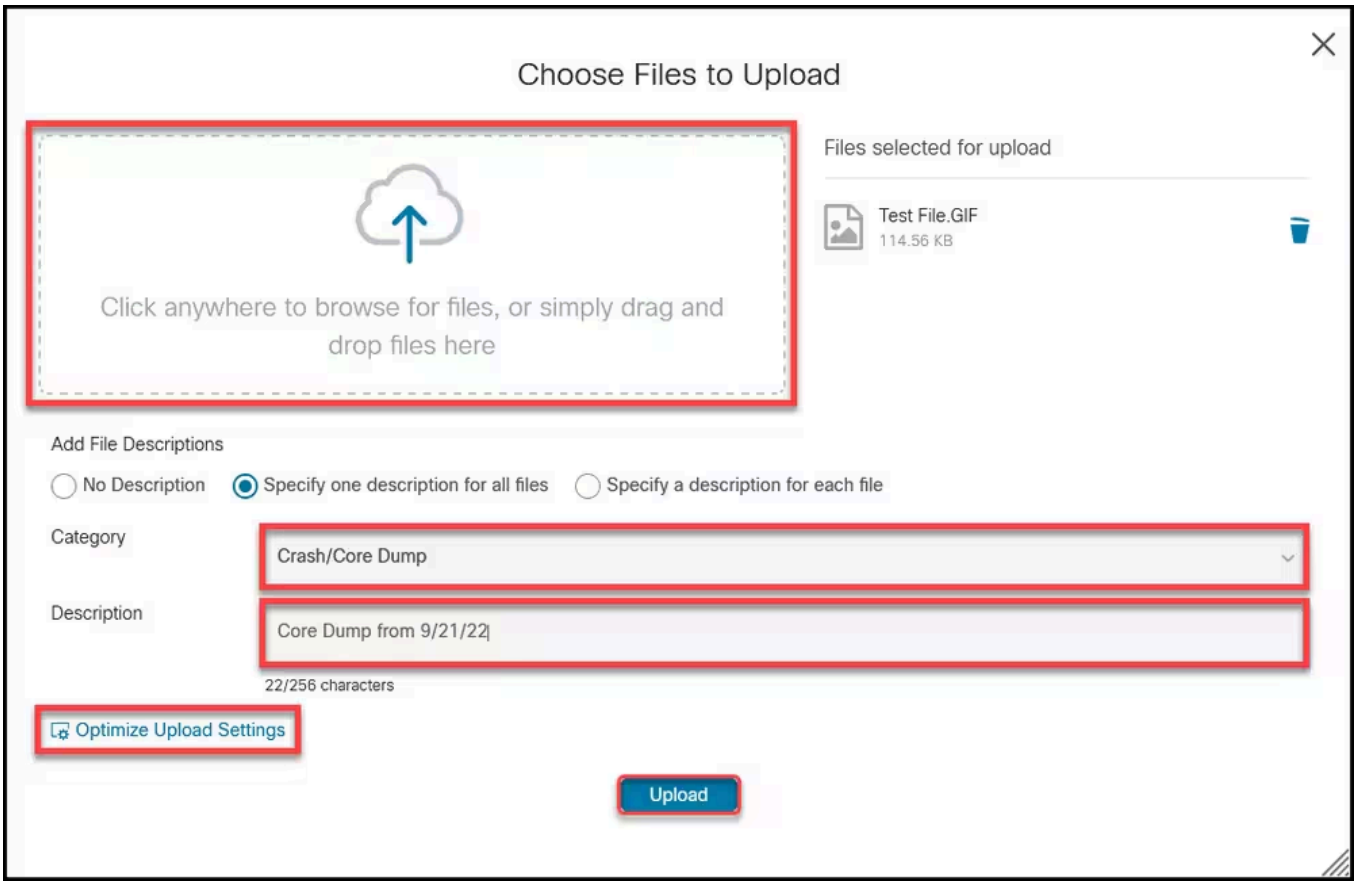


Step 4. In the **Choose Files to Upload** dialog box, drag the files that you want to upload or click inside to browse your local machine for files to upload.

Step 5. Add a description and specify a category for all files, or individually.

> **Note:** In order to optimize the upload settings for your network condition, click **Optimize Upload Settings**.

Step 6. Click **Upload** in order to start the upload process.



Step 7. Once all uploads are complete, you can close the window or click **Add More Files** in order to upload

more files.



Step 8. Uploaded files can be managed in the **Attachments** tab.



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Customer eXperience Drive

Service Summary

The Customer eXperience Drive (CXD) is a multi-protocol file upload service with no limitation on the uploaded file size. It helps Cisco customers with active Service Requests (SRs) to upload data directly to a case using a unique set of credentials created per SR. The protocols supported by CXD are natively supported by Cisco products which allows for uploading directly from Cisco devices to SRs.

Supported Protocols

Table 2 summarizes the protocols supported by CXD. It is worth noting that regardless of the protocol used, there is no limit set on the uploaded file size.

Table 2. CXD Supported Protocols

Name	Protocol/Port	Encrypted	Data Channel Ports	Notes
Secure File Transfer Protocol (SFTP)	TCP/22	Yes	N/A	
Secure Copy Protocol (SCP)	TCP/22	Yes	N/A	
Hyper Text Transfer Protocol	TCP/443	Yes	N/A	Only API-based uploads are

over SSL (HTTPS)				supported.
File Transfer Protocol of SSL (FTPS) Implicit	TCP/990	Yes	30000-40000	Firewalls cannot inspect FTPS, as the control channel is encrypted. Hence, the firewall needs to allow outbound connectivity to the entire data channel port range.

CXD Upload Token

CXD creates unique upload tokens per SR. The SR number and the token are used as the username and password to authenticate to the service and subsequently upload files to the SR.

> **Note:** The token is for upload only and does not allow the user to access case files, or even files currently being uploaded. If the user would like to view case files, that can only be done in SCM.

Retrieving the Upload Token for an SR

Using SCM

When an SR is opened, users must create the upload token to upload the attachment.

In order to retrieve/generate the upload token, complete these steps:

Step 1. Log in to SCM.

Step 2. In order to view and edit a case, click the case number or case title in the list. The Case Summary page opens.

Step 3. Click the **Attachments** tab.

Step 4. Click **Generate Token**. Once the token is generated it is displayed next to the Generate Token button.

> **Note:** The username is always the SR number. The terms password and token refer to the upload token, which is used as a password when prompted by CXD.

Uploading Files to CXD

Using Desktop Clients

In general, all the user needs to do is use a client, depending on the protocol, to connect to cxd.cisco.com, authenticate using the SR number as the username and the upload token as the password, and eventually upload a file. Depending on the protocol and the client, the user steps can be different. It is always recommended to refer to the client documentation for more details.

Directly from a Cisco Device

All Cisco devices have built-in file transfer clients, usually utilized using a **copy** or **redirect** command. Cisco equipment running on a Linux distribution usually supports one or more of scp, sftp, and curl for SCP, SFTP, and HTTPS integrations.

File Upload API

The file upload API utilizes the HTTP PUT verb to upload files to CXD. For maximum compatibility and simplicity of integration, the API is kept simple.

HTTP Method: PUT

URL: https://cxd.cisco.com/home/<*destination file name*>

Headers:

Table 5. CXD File Upload API Headers

Key	Type	Value
Authorization	String	Basic HTTP Auth String

The body is the file data itself. There are no fields or forms here, which makes the request very simple.

Sample Python Code to use the PUT API

Note that the code assumes that the file is stored in the same path you are running it from.

```
import requests
from requests.auth import HTTPBasicAuth

username = 'SR Number'
password = 'Upload Token'
auth = HTTPBasicAuth(username, password)

filename = 'showtech.txt'  # Destination filename
url = f'https://cxd.cisco.com/home/{filename}'

headers = {"Expect": "100-continue"}

file_path = 'Local Path to the File'

with open(file_path, 'rb') as f:
    r = requests.put(url + filename, f, auth=auth, headers=headers)
    if r.status_code == 201:
        print("File Uploaded Successfully")
```

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Email File Attachment Uploads

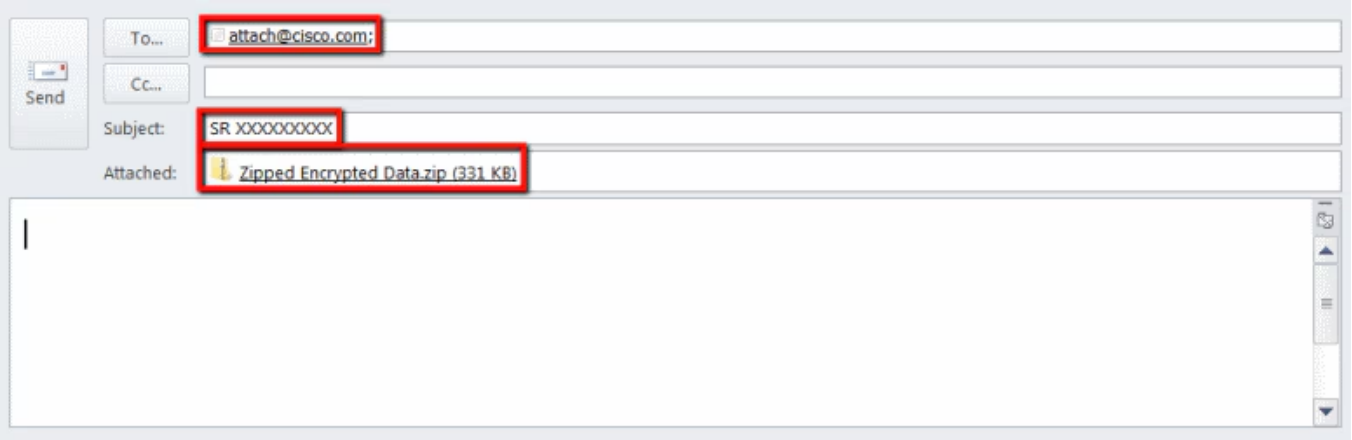
If SCM and CXD do not work for you, another alternate file upload method is email file attachment upload. Note that this method is *fundamentally insecure* and does not encrypt the file or the communication session used to transport the file between the customer and Cisco. It is incumbent upon the customer to explicitly encrypt files before the files are uploaded through email file attachments. As an additional security best

practice, any sensitive information such as passwords need to be obfuscated or removed from any configuration file or log that is sent over an insecure channel. For more information, see Encrypting Files.

After the files are encrypted, upload additional information and files to the case by sending the information via an email message to attach@cisco.com with the case number in the subject line of the message, for example, subject = Case xxxxxxxxx.

Attachments are limited to 20 MB per email update. Attachments submitted by using email messages are not encrypted in transit, but are immediately linked to the specified case and stored in an encrypted format.

Attach the file to an email message and send the message to attach@cisco.com as shown in this screenshot.



The previous screenshot displays a Microsoft Outlook email that has an encrypted ZIP file attachment, the correct To address, and a properly formatted Subject. Other email clients need to provide the same functionality and perform just as well as Microsoft Outlook.

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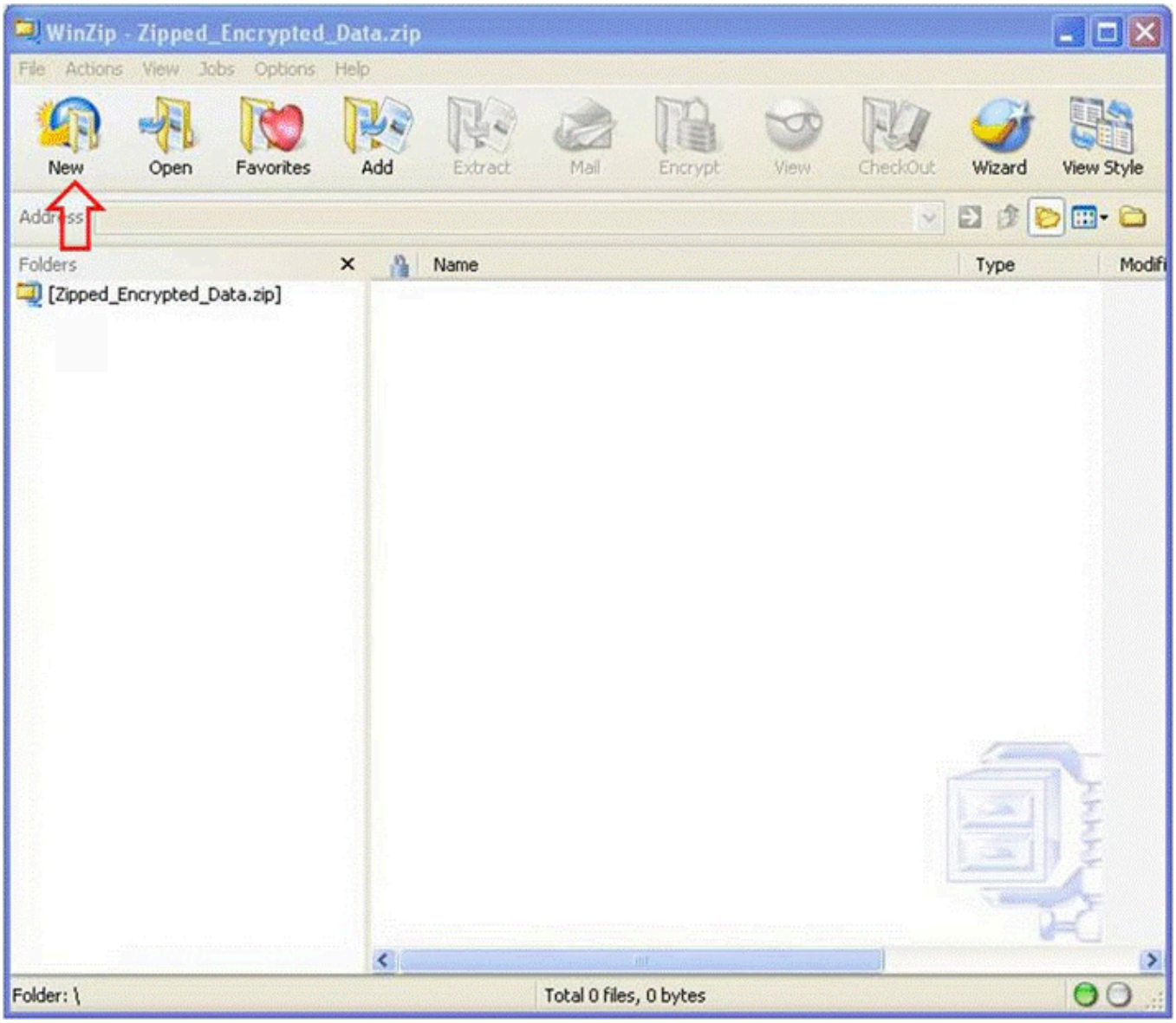
Encrypting Files

The following examples show how to encrypt files by using three of the many available options such as WinZip, Linux tar and openssl commands, and Linux Gzip and GnuPG. A strong encryption cipher such as AES-128 needs to be used in order to properly protect the data. If you are using ZIP, an application that supports AES encryption must be used. Older versions of ZIP applications support a symmetric encryption system that is not secure and is not to be used.

Encrypting Files Using WinZip

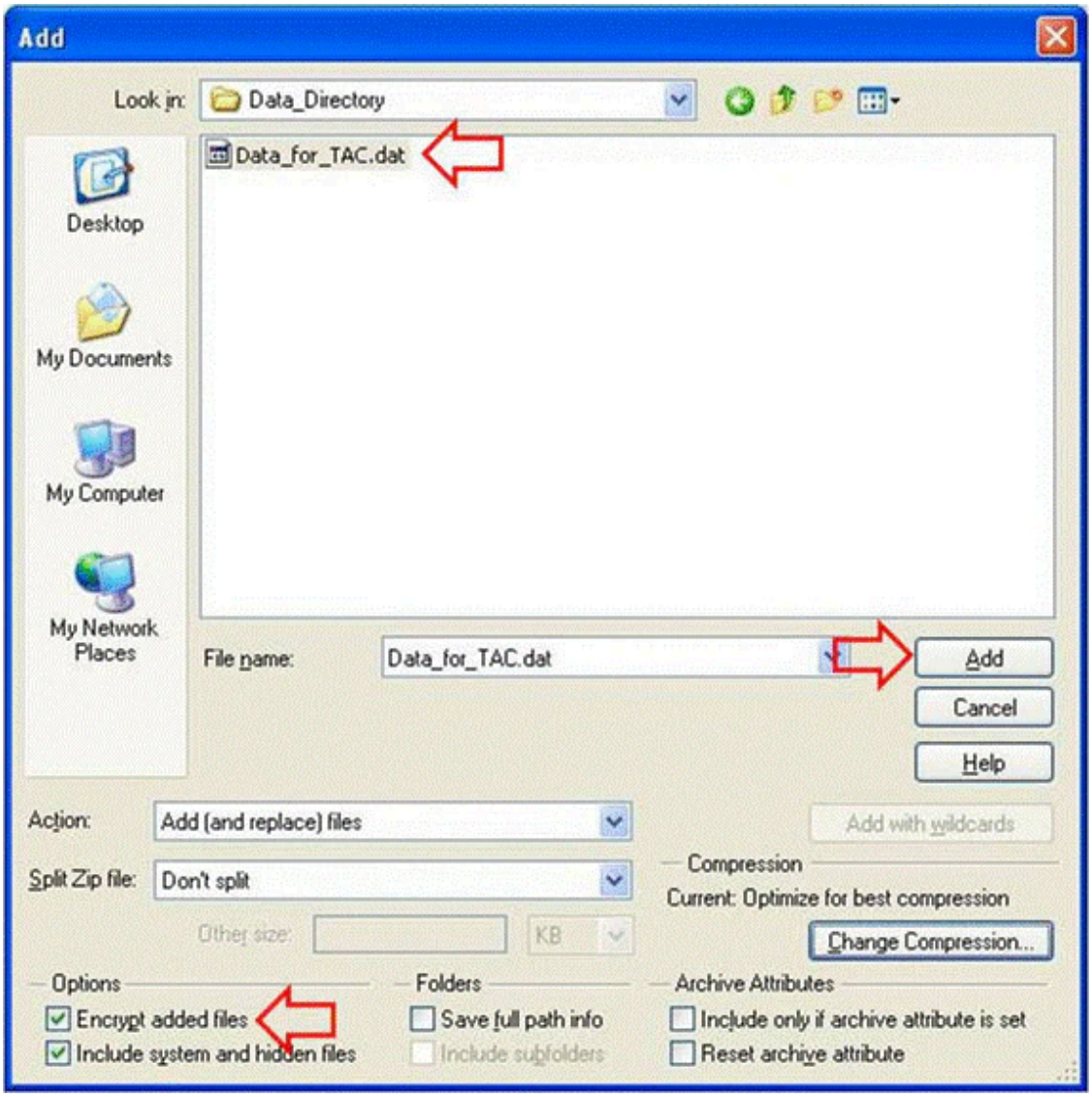
This section demonstrates how to encrypt files by using the WinZip application. Other applications provide the same functionality and perform as well as WinZip.

Step 1. Create a ZIP archive file. In the WinZip GUI, click **New** and follow the menu prompts in order to create an appropriately named, new ZIP archive file. The system displays the newly created ZIP archive file.



Step 2. Add the file(s) to be uploaded to the ZIP archive file and check the **Encrypt added files** check box. From the main WinZip window, click **Add** and then choose the file(s) to upload. The **Encrypt added files**

check box must be checked.



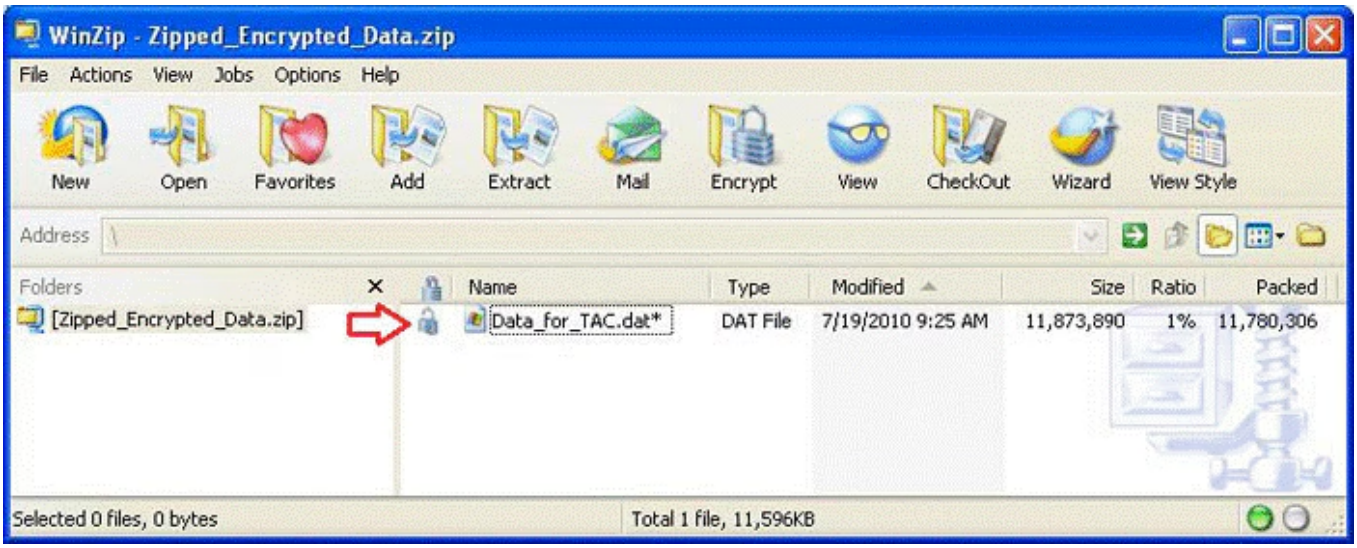
Step 3. Encrypt the file by using the AES encryption cipher and a strong password:

1. Click **Add** in the file selection window in order to open the **Encrypt** window.
2. In the **Encrypt** window, create an appropriately strong password. The password is shared with the case Customer Support Engineer owner, as discussed in Communicating the Password to the TAC Customer Support Engineer.
3. Choose one of the AES Encryption methods.

4. Click **OK** in order to encrypt the file(s) and display the main WinZip window.



Step 4. Verify that the file is encrypted. Encrypted files are marked with an asterisk following the file name or a lock icon in the Encryption column.



Encrypting Files Using Tar and OpenSSL

This section shows how to encrypt files by using the Linux command-line **tar** and **openssl** commands. Other archive and encryption commands provide the same functionality and perform just as well under Linux or Unix.

Step 1. Create a tar archive of the file and encrypt it through OpenSSL using the AES cipher and a strong password as demonstrated in the following example. The command output shows the combined **tar** and **openssl** command syntax to encrypt the file(s) using the AES cipher.

```
[user@linux ~]$ tar cvzf - Data_for_TAC.dat | openssl aes-128-cbc -k
Str0ng_passWo5D |
    dd of=Data_for_TAC.aes128 Data_for_TAC.dat
```

60+1 records in
60+1 records out

Encrypting Files Using Gzip and GnuPG

This section demonstrates how to encrypt files by using the Linux command-line Gzip and GnuPG commands. Other archive and encryption commands provide the same functionality and perform just as well under Linux or Unix. The command output shows how to use the gzip and gpg command syntax to encrypt the file(s) using the AES cipher.

Step 1. Compress the file by using Gzip:

[user@linux ~]$ gzip -9 Data_for_TAC.dat

Step 2. Encrypt the file through GnuPG using the AES cipher and a strong password:

```
user@linux ~]$ gpg –cipher-algo AES –armor –output Data_for_TAC.dat.gz.asc
–symmetric Data_for_TAC.dat.gz
```

Step 3. Enter and confirm the strong password at the passphrase prompt:

Enter passphrase:
Repeat passphrase:

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Communicating the Password to the TAC Customer Support Engineer

When encrypting attachments, share the encrypting password with the case Customer Support Engineer owner. As a best practice, use a method other than the one used to upload the file. If you used an email message or FTPS to upload the file, communicate the password out-of-band such as by telephone or SCM case update.

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Customer Case Attachments Data Management

Files uploaded to the **CXD folder** are managed in accordance with Cisco's policies. Documentation regarding data controls and retention can be found in the TAC Privacy Data Sheet.

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Summary

Multiple options exist for uploading information to TAC to help them resolve cases. SCM offers secure uploads through a browser, while CXD supports uploads through a browser, Web API, and multiple protocols compatible with different types of clients and Cisco devices. If you cannot use SCM or a CXD-supported protocol other than FTP for your file upload, the least preferred option is to send an email message to attach@cisco.com. If you use either of these options, it is strongly advised that you encrypt your files before transit. For more information, see Encrypting Files. You need to employ a strong password and communicate the password to the case Customer Support Engineer out-of-band such as by telephone or SCM case update.

For the duration that a case is open and for a period up to 18 months following final closure of a case, all files are instantly accessible from within the case tracking system to authorized Cisco personnel.

- After 18 months the files can be moved to archival storage.
- At any time, an authorized customer contact can expressly request that a specific file be purged from a case.

- Files in the **CXD folder** are retained for only four days.

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Additional Information

- Accessing Cisco Technical Services
- Cisco Worldwide Support Contacts
- Cisco Technical Services Resource Guide
- Cisco Conferencing Products
- The GNU Privacy Guard
- The OpenSSL Project
- WinZip

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Revision History

Revision	Publish Date	Comments
3.0	12-Apr-2024	Removed the Customer File Retention section and added the Customer Case Attachments Data Management section.
2.0	01-Nov-2023	Removed the Case File Uploader section and updated the Customer eXperience Drive section.
1.0	14-Nov-2017	Initial Release



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